

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

Summary & Protocols Used: The `tcpdump` log analysis captures network communication between client `192.51.100.15` and the DNS server `203.0.113.2`. The network traffic involves two main Layer 4 protocols: **UDP** (User Datagram Protocol) for the outgoing DNS query, and **ICMP** (Internet Control Message Protocol) carrying the returning diagnostic error message.

Log Details & Trends:

- **DNS Request Attempts:** The client machine (`192.51.100.15.52444`) repeatedly attempts to resolve the IP address for the website `yummyrecipesforme.com` by sending standard DNS queries (`35084+ A? yummyrecipesforme.com.`) to destination port 53 on server `203.0.113.2` at `13:24:32`, `13:26:32`, and `13:28:32`.
- **Error Response:** Instead of a valid DNS answer packet containing an IP address, the destination server (`203.0.113.2`) immediately responds with an ICMP error packet: `ICMP 203.0.113.2 UDP port 53 unreachable`.
- **Repetition:** The client retries the request every two minutes, and each attempt is met with the exact same ICMP port unreachable response.

Interpretation of the Issue:

- **Port Role:** Port 53 is the standard well-known port for the Domain Name System (DNS), which translates human-readable domain names into IP addresses.
- **Root Cause Assessment:** The ICMP Type 3 error indicates that while host `203.0.113.2` is reachable at the IP layer, no active process or

daemon is listening on UDP port 53. The most likely cause is that the DNS service has crashed, the server is misconfigured, or a firewall/security control is actively rejecting inbound UDP traffic on port 53. Consequently, client machines cannot resolve yummyrecipesforme.com and web browsing fails.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Part 2: Explain your analysis of the data and provide at least one cause of the incident

- **Time Incident Occurred / First Reported:** The incident was first logged at **13:24:32**, with recurring query failures and ICMP error packets continuing through **13:28:50**.
- **Scenario, Events, and Symptoms:** Users reported an inability to access the external website yummyrecipesforme.com via their web browsers. The primary symptom is a complete failure to resolve domain names to IP addresses on the local network.
-
- **Current Status of the Issue:** Ongoing; client machines remain unable to access domain-based web resources through the configured DNS server at 203.0.113.2.
- **Information Discovered During Investigation:**
 1. Network packet capture using [tcpdump](#) confirms client [192.51.100.15](#) transmits standard DNS queries ([A? yummyrecipesforme.com.](#)) over UDP to destination port 53 on [203.0.113.2](#).
 2. The destination server [203.0.113.2](#) consistently responds with [ICMP 203.0.113.2 udp port 53 unreachable](#) error packets.
 3. Layer 3 (IP) connectivity is active, but Layer 4 (UDP Port 53) communication is rejected, preventing any DNS records from being returned.

4.

- **Suspected Root Cause:** The DNS daemon/service on host 203.0.113.2 has crashed or stopped running, or a recently updated firewall rule is actively rejecting inbound UDP traffic on port 53.
- **Next Steps & Solution to Implement:**
 1. Verify the service status of the DNS daemon (e.g., `systemctl status named` or `bind9`) on 203.0.113.2 and restart the service if inactive.
 2. Inspect local and network firewall configurations (such as `iptables` or perimeter access control lists) to verify inbound UDP traffic on port 53 is permitted.
 3. Configure a temporary fallback secondary DNS resolver (e.g., 8.8.8.8 or 1.1.1.1) on client machines or DHCP scope to restore web access immediately while server remediation takes place.